

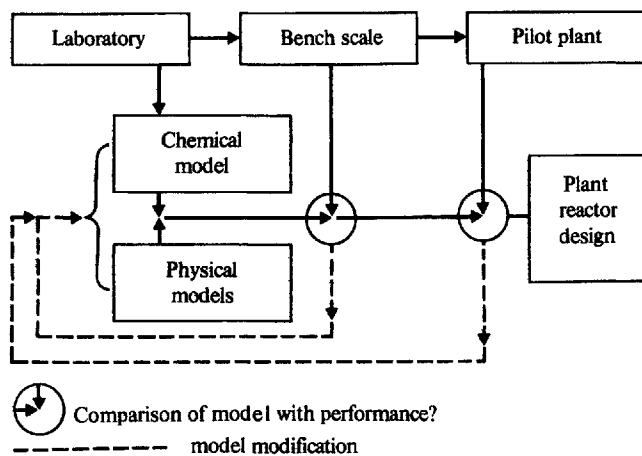


FIGURE 1-3
Realistic reactor-design flowsheet.

Between the two extreme modes of inquiry (the dangerous rational-synthesis approach and the cautious but expensive incremental scale-up strategy) some prudent intermediate mode exists. We might schematize this mode as shown in Fig. 1-3.

While Fig. 1-3 may seem more complex than the alternatives, the dangers of the ideal approach and costs of the incremental-scale-up strategy must be considered if, as is not uncommon, we are confronted with a 1000:1 or 10,000:1 scale-up factor between commercial plant and test tube. The intermediate alternative scheme involves *feedback*; i.e., one attempts analysis at laboratory and bench scale, and then a comparison of pilot-plant prediction and performance is made. This confrontation of reality with predictions rooted in physicochemical model prediction allows for model-parameter (physical and chemical) adjustment. Therefore, final design is not an a priori process but one that is informed through feedback.

In sum, the chemical reaction engineer is more likely to be confronted with problems of analysis of existing units (laboratory, bench, pilot, and plant scale) than with the ideal issue of a priori plant-reactor design. It follows that the more intelligent we become in analyzing at any scale of operation, the fewer scales of operation will have to be analyzed. Returning to the SO_2 oxidation example, the more likely route to final plant design will involve (1) combining isothermal laboratory data with physical, heat- or mass-transfer data, (2) comparing predictions based upon a comprehensive physicochemical model with nonisothermal bench or semiworks data, (3) modifying model components to account for discrepancies in step (2), and (4) repeating steps (2) and (3) at larger scales of operation.

Implicit in the above discussion is the notion that with an increase in scale of reactor operation (size) certain physical events intervene to alter, modify, and indeed possibly falsify chemical-kinetic dispositions. In the simple case of a

homogeneous reaction, it may well be that heat transport rather than chemical reaction per se determines reactor size and/or its performance. In a heterogeneous reaction, since, by definition, at least two phases are involved, mass and heat transport between phases as well as with the external environment may be involved. The emphasis in this text lies on heterogeneous reactions, as such reactions are, in the author's opinion, more common and far more challenging.

A discussion of an elementary example should suffice to illustrate the intervention of the physical upon chemical events with respect to the chemical reaction.

1-1 THE REACTION

An important distinction must be made between the intrinsic rate of a chemical reaction and the observed, or global, rate of reaction. An elementary example will suffice to illustrate this point. Let us suppose that an inert gas such as He, containing a few mole percent NH_3 , is passed over a hot tungsten wire. The tungsten filament catalyzes the decomposition of NH_3 . The overall reaction is $2\text{NH}_3 \xrightarrow{W} \text{N}_2 + 3\text{H}_2$.

The situation can be schematically set forth as shown in Fig. 1-4. The intrinsic rate of the chemical reaction will be proportional to the NH_3 concentration at the catalyst surface C_s . Assuming for the sake of illustration that this surface rate is directly proportional to C_s ,

$$\text{Intrinsic rate} = k_s C_s \quad (1-1)$$

where k_s is the intrinsic chemical-reaction-rate coefficient. Now in the steady-state circumstance the intrinsic rate must be equal to the rate at which NH_3 is supplied to the surface via gas-film mass transport. In the traditional manner we designate the mass-transport coefficient $k_g a$. Thus

$$k_g a (C_0 - C_s) = k_s C_s \quad (1-2)$$

Mass transport = surface reaction

Consider next the problem of analyzing the above experiment; i.e., how do we determine the intrinsic coefficient k_s ? As our filament is small, NH_3 conversion

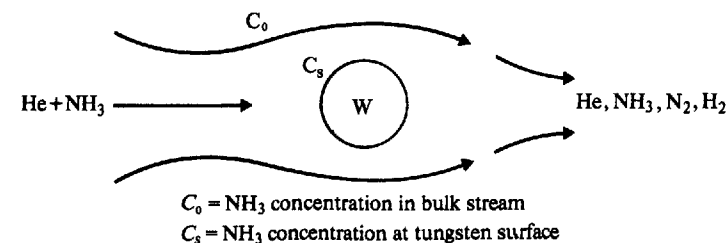


FIGURE 1-4
A nonporous-solid-catalyzed reaction-flow network.

will be small indeed, and we can assume an average value of reactant concentration based upon chemical analyses up- and downstream of the tungsten wire. But such measurements provide C_0 not C_s ; for the surface concentration is usually unobservable. So our observed rate is given by

$$\text{Observed rate} = k_0 C_0 \quad (1-3)$$

Clearly k_0 does not equal k_s , the coefficient sought, unless, as Eq. (1-2) shows, $C_0 = C_s$; that is, $k_g a$ is very large and so provides a supply of NH_3 to the surface at a rate which does not cause a gradient in the gas film. Quantitatively we solve Eq. (1-2) for C_s

$$C_s = \frac{C_0}{1 + k_s/k_g a} \quad (1-4)$$

and substitute Eq. (1-4) into (1-1):

$$\text{Intrinsic rate} = k_s C_s = \frac{k_s C_0}{1 + k_s/k_g a} \quad (1-5)$$

Bearing in mind that our observed rate must equal the intrinsic rate (note that observed and intrinsic rate *coefficients* are equal only when $C_s = C_0$), we see that

$$k_0 C_0 = k_s C_s$$

$$\text{or} \quad k_0 C_0 = \frac{k_s C_0}{1 + k_s/k_g a} \quad (1-6)$$

and so our observed, or *global*, rate coefficient is in general related to the *intrinsic* coefficient (for linear kinetics) by

$$k_0 = \frac{1}{1/k_s + 1/k_g a} \quad (1-7)$$

an expression reminiscent of that for the overall heat-transfer coefficient in terms of individual coefficients.

Had our reaction been noncatalytic, say thermal decomposition of NH_3 , global and intrinsic rate coefficients would, of course, be identical (assuming the absence of free-radical diffusion intrusions).

What has been noted above concerning mass-diffusional masking of intrinsic kinetics can also be said of the diffusion of heat. That is, one must also anticipate that a bulk-fluid temperature may differ, by reason of film resistance to heat transfer, from that at the reaction site (in the above example, the tungsten surface).

In sum

$$\text{Global rate} = k_0 f(C_0, T_0) \quad \text{and} \quad \text{intrinsic rate} = k_s f'(C_s, T_s)$$

In the instance of homogeneous reaction we can generally be confident that $k_0 = k_s$. For a heterogeneous system, one would be ill advised to make such assumptions a priori.

This text treats first the intrinsic events and then explicitly heterogeneous systems in terms of global rates and their physicochemical components.

1-2 THE REACTOR

The device within which the physicochemical transformations are caused can assume various shapes and modes of operation and be operated in a number of possible environments of temperature and pressure. Given some knowledge of the chemistry and rates of the various reactions involved, the reactor environment which provides the desired quantity of product at an acceptable level of quality must be specified. Just as keen competence is required in analyzing and formulating quantitative models of the chemical reactions, so that an equally profound awareness of reactor behavior is required of the chemical reaction engineer.

1-3 STRUCTURE OF THIS TEXT

Given the key importance of physicochemical kinetics and reactor behavior, this text is structured to:

- 1 Review principles and techniques whereby models of intrinsic rates of chemical reactions can be fashioned (Chap. 2)
- 2 Introduce key concepts of reactor behavior in terms of limiting reactor types and environments (Chap. 3)
- 3 Set forth governing continuity equations for the nonideal reactor, thereby identifying the nature and magnitudes of real reactor parameters, with design and analyses of simple homogeneous reactor types (Chap. 4)
- 4 Introduce general concepts underlying global heterogeneous reactions (Chap. 5)
- 5 Treat specific noncatalytic heterogeneous systems, both fluid-fluid and fluid-solid (Chaps. 6 and 7)
- 6 Review principles of heterogeneous catalysis and intrinsic rate formulations (Chap. 8)
- 7 Treat the global rates of heterogeneous catalytic reactions (Chap. 9)
- 8 Outline principles of design and analysis of common heterogeneous reactor types (Chap. 10)

1-4 THE ANATOMY OF PROCESS DEVELOPMENT

An appropriate framework within which the issues treated in this text can fruitfully be focused is provided by Weekman.¹ In Fig. 1-5 the flowsheet between exploratory studies and plant operation is indicated. Note the early intervention of

¹ V. W. Weekman, course notes on heterogeneous catalysts, University of Houston.

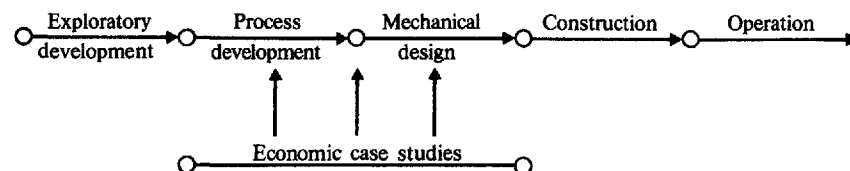


FIGURE 1-5
Scale-up flowsheet.

economic analyses. In greater detail Fig. 1-6 reveals the role of kinetic-transport experiments, model development, and optimization.

If we suppose that yield-conversion behavior is to be modeled, Fig. 1-7 illustrates such modeling on three self-explanatory levels of sophistication. Note the hazards of empirical modeling in Fig. 1-7a and b when extrapolation is required.

A scale whereby models can be rated is presented in Fig. 1-8, where the barometer, or index, of fundamentalness is the ratio of the number of fundamental laws invoked to the number of adjustable constants. Diverse petroleum-refining reactor systems are indicated on this scale.

Prater's principle of "optimum sloppiness" is schematized in Fig. 1-9, where usefulness, cost, and *net* value of the model are plotted against the index of fundamentalness. This display shows the unreasonableness of seeking a totally fundamental model of a complex reactor. Such a reactor hosts a complex array of physicochemical rate phenomena, the fundamentals of which are ill understood and thus proper subjects of long-range fundamental research programs, usually defying the time table of process development and plant design. We may confidently assert, however, that a command of fact and theory will reduce the cost of model creation with evident net value benefit.

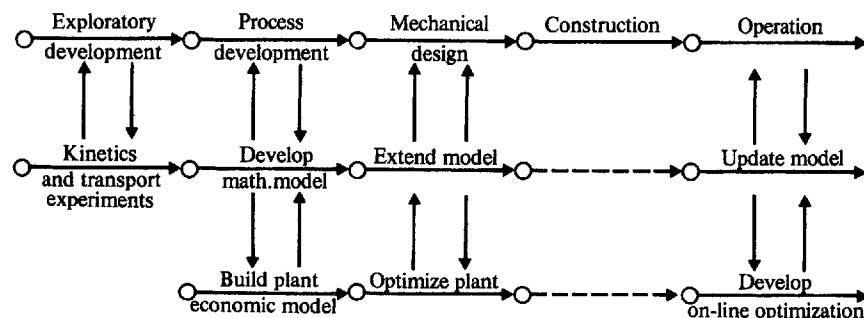


FIGURE 1-6
A somewhat more detailed scale-up flowsheet.

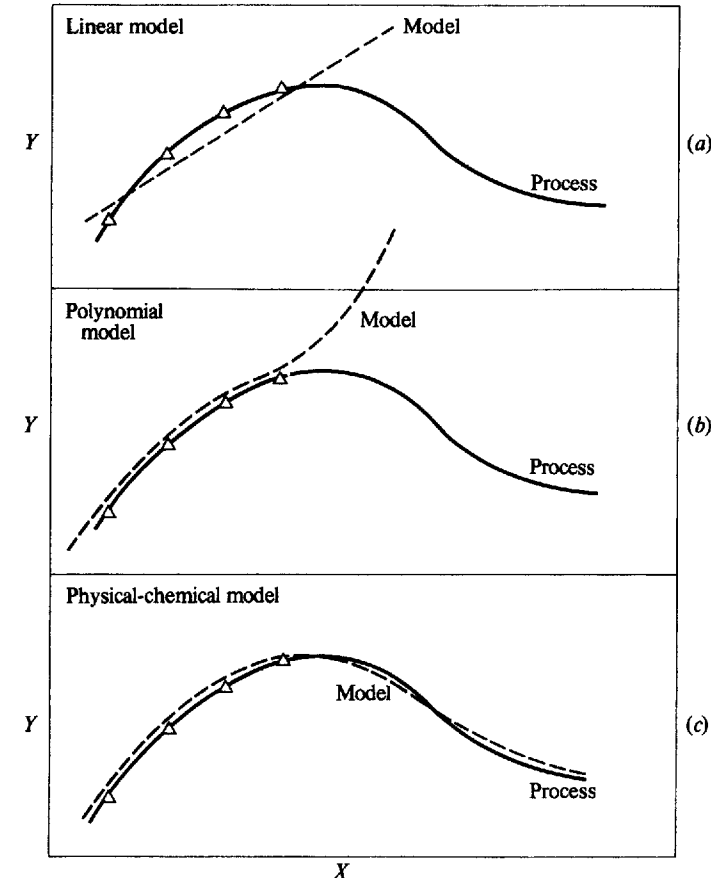


FIGURE 1-7
Diverse extrapolation models.

Rating of models

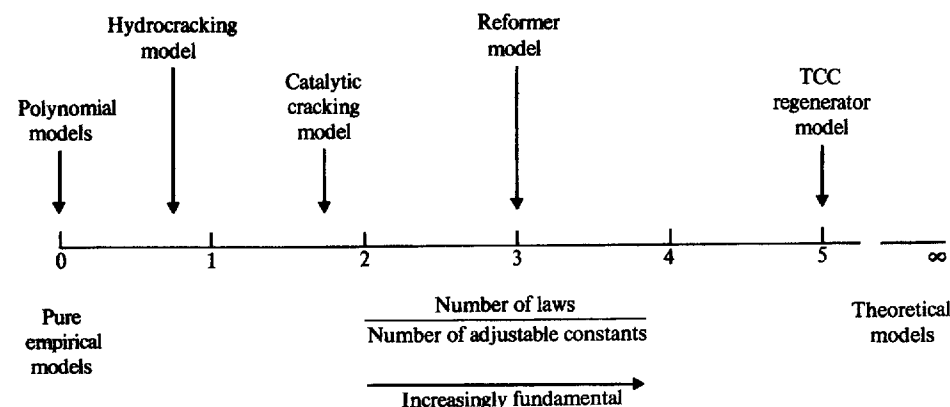


FIGURE 1-8
Model ratings in terms of the index of fundamentalness.

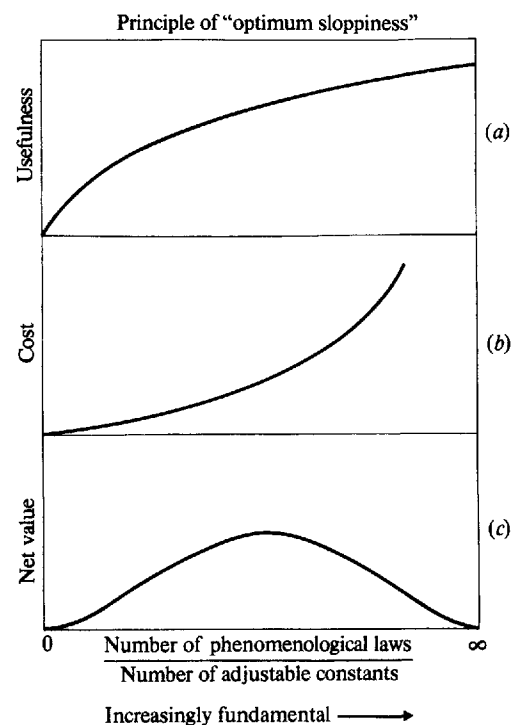


FIGURE 1-9
Prater's principle of optimum sloppiness.

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"The velocity is delightful . . ." "The Greville Memoirs," July 18, 1837

Introduction

In this chapter the experimental bases of chemical-reaction-rate expressions are noted, and definitions are set forth in terms of laboratory observables. Chemical-reaction-rate theories are briefly discussed insofar as such discussion establishes some rational bases in support of laboratory-rooted definitions. A brief comment upon limiting laboratory reactor types is then presented, with some necessary definitions designed to facilitate appreciation of the formal modes of rate analysis. Formal, mathematical descriptions of chemical-reaction rates are presented for both simple and complex reaction schemes, and various means of data analysis are implicitly suggested, with particular emphasis upon yield and selectivity as well as conversion. The chapter concludes with a brief treatment of autocatalytic and chain reactions and some remarks on data procurement and error analyses. This chapter, then, is designed as a review of matter usually found in an undergraduate physical chemistry course. An added emphasis is evident in that rate of generation of product (yield) rather than mere disappearance of reactant is given special attention.

Homogeneous, isothermal, constant-volume reaction environments are assumed to prevail in the analyses presented in this chapter. Such an atmosphere

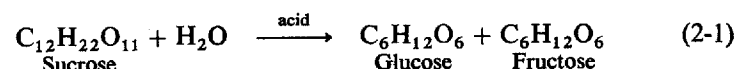
must be striven for if meaningful kinetic data are to be extracted from a laboratory reactor. The more complex systems (heterogeneous and/or variable-volume) will be treated in due course. At this juncture, a respect for simple definitions and a familiarity with the general tenets of kinetic analysis seem to take highest priority.

2-1 EXPERIMENTAL BASES OF CHEMICAL KINETICS

Whereas the concern of chemical equilibrium or statics is that of specifying *what* reactions may occur between molecular and/or atomic species, chemical kinetics concerns itself with the *velocity* of reactions between species. Application of chemical-equilibria principles indicates, for example, that in the oxidation of NH_3 reactions may occur to yield N_2 , NO , and N_2O as well as H_2O . If interest is in one product, say NO , it is clear that thermodynamics simply assures us that said product can, in principle, be produced. Whether NO can actually be produced and produced to the exclusion of N_2 and N_2O will depend upon the kinetics of the various reactions. If conditions of temperature, pressure, and appropriate catalysis can be found, the rate of NO formation will perhaps be relatively rapid compared with N_2 and N_2O generation. In fact in the presence of a Pt-alloy catalyst at 1000°C and 1 atm pressure, nearly 100 percent yield of NO can be obtained from a mixture of air and about 10 percent NH_3 . On the other hand, at temperatures below 600°C , N_2O formation predominates.

Given the obvious importance of chemical kinetics, our concern is now that of defining and specifying how the velocity of a chemical reaction is expressed. Unlike chemical statics, precise chemical-kinetic principles and data do not exist to permit a priori prediction. Chemical kinetics rests largely upon experiment.

In 1850, L. Wilhelmy made the first *quantitative* observations of reaction velocity. In his batch study of sucrose inversion in aqueous solutions of acids,



Wilhelmy noted that the rate of change of sucrose concentration C with respect to time t is a linear function of the concentration of unconverted sucrose C . That is,

$$-\frac{dC}{dt} = kC \quad (2-2)$$

The observed velocity law reveals that reaction (2-1) is irreversible; i.e., the back reaction between glucose and fructose is negligible relative to the forward sucrose-inversion reaction.

The proportionality constant k is known as the *specific rate coefficient*. Actually for reaction (2-1), k is proportional to acid concentration. The acid is clearly a catalyst in the reaction, as it does not appear in the overall reaction statement.

At a fixed temperature and acid concentration, Eq. (2-2) was integrated by Wilhelmy to yield, for an initial concentration C_0 ,

$$\ln \frac{C}{C_0} = -kt \quad \text{or} \quad \frac{C}{C_0} = \exp[-kt] \quad (2-3)$$

The concentration-vs.-time data agreed nicely with the above equation.

In the system studied by Wilhelmy, the rate, being proportional to the first power of concentration, is said to be of *first order* in the concentration of that species. *Order of reaction* with respect to a particular species is the numerical value of the power to which the concentration is raised to faithfully describe the experimental relationship between reaction rate and the concentration of that species; i.e.,

$$\text{Rate} = -\frac{dA}{dt} = kA^a \quad (2-4)$$

The overall reaction order is the sum of exponents; i.e., if

$$-\frac{dA}{dt} = kA^a B^b C^c \quad (2-5)$$

then the overall order is $\alpha + \beta + \gamma$.

Essentially order is empirical insofar as it is specified on the basis of observed rate-vs.-concentration data. Order need not be a whole number, since it may be zero, fractional, or negative for a specific component.

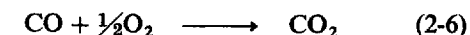
Thus in the sucrose-inversion reaction, we can say that the rate is first order with respect to sucrose and zero order with respect to the coreactant, water. Zero order with respect to water does not imply that the reaction does not involve water but that there is no apparent *experimental* dependence since water was present in such vast excess that its concentration changed negligibly during reaction. That is,

$$\text{Rate} = k(\text{sucrose})^1(\text{H}_2\text{O})^0 = k'(\text{sucrose})^1$$

as $(\text{H}_2\text{O})^0$ is a constant in the experiment. In principle, the rate is β th order in water, and this dependency would be observed experimentally if H_2O were not present in vast excess.

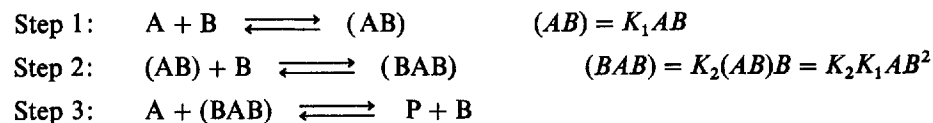
Order and Stoichiometry

Essentially order is determined by the best fit or correspondence between a rate equation and experimental data. It follows that there is no necessary connection between kinetic order and the stoichiometry of the reaction. For example, in the Pd-catalyzed oxidation of CO , the stoichiometry is

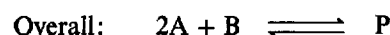


while kinetics suggest *negative* first order in CO . Equation (2-6) is simply an overall statement of the reaction, whereas the process must proceed through a

series of more elementary steps. The sum of these elementary processes must yield the overall reaction statement. However, in a kinetic study, one necessarily measures only the slowest step or steps of the sequence. More formally the rate-determining step is measured implicitly, and consequently its concentration dependencies will appear in the rate expression. The resultant rate expression may be complex or simple. In either case, it is not possible to infer a mechanism of reaction (detailed sequence of steps) solely from kinetic rate expressions. For example, let an overall reaction statement be $2A + B \rightleftharpoons P$. Suppose the elementary steps (mechanism) to be:



where K_1 and K_2 are equilibrium coefficients.



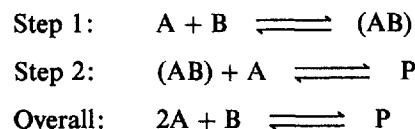
The following irreversible rate laws can be found:

If step 1 is the slowest, rate = $k_1 AB$.

If step 2 is the slowest, rate = $k_2 K_1 AB^2$.

If step 3 is the slowest, rate = $k_3 K_2 K_1 A^2 B^2$.

Suppose on the other hand, the mechanism is



Then if step 1 controls, rate = $k_1 AB$, while if step 2 controls, rate = $k_2 K_1 A^2 B$, where k_1 and k_2 are rate coefficients. These rate expressions are derived as follows:

In the first illustration, when a step is assumed to be rate-controlling, i.e., the slowest of all steps, all other elementary steps are assumed to be in a state of rapidly established equilibrium, or steady state. Then if step 1 in the first illustration controls, we can write the rate of that reaction as indicated by the stoichiometry of that one elementary step; i.e.,

$$\text{Rate (step 1)} = k_1 AB$$

Should step 2 be the slowest elementary event in the sequence (mechanism), then

$$\text{Rate (step 2)} = k_2 (AB)B$$

but by the equilibrium in step 1

$$K_1 = \frac{(AB)}{AB} \quad \text{or} \quad (AB) = K_1 AB$$

Hence

$$\text{Rate (step 2)} = k_2 (AB)B = k_2 K_1 AB^2$$

When step 3 controls,

$$\text{Rate (step 3)} = k_3 A(BAB)$$

but

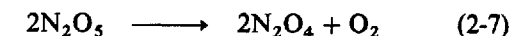
$$(BAB) = K_2 (AB)B = K_2 K_1 AB^2$$

Therefore

$$\text{Rate (step 3)} = k_3 K_2 K_1 A^2 B^2$$

Some concrete examples regarding order and stoichiometry, model and mechanism can be cited.

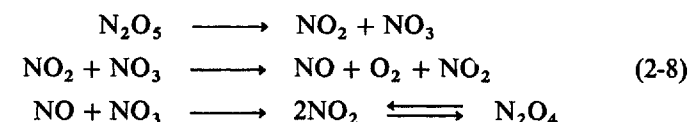
Kinetic studies of the decomposition of N_2O_5



indicate

$$-\frac{d(N_2O_5)}{dt} = k(N_2O_5)$$

It is incorrect to infer that reaction (2-7), while first order, involves simple (unimolecular) dissociation of N_2O_5 to final products N_2O_4 and O_2 . Ogg¹ showed that the mechanism is



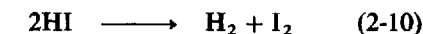
Consider the decomposition of acetaldehyde, $CH_3CHO \rightarrow CH_4 + CO$, in the gas phase at 450°C. The rate expression is

$$-\frac{d(CH_3CHO)}{dt} = k(CH_3CHO)^{3/2} \quad (2-9)$$

Clearly order and stoichiometry are not synonymous. Further, as molecules react as integers (mono- or bimolecular), the observed rate law cannot represent a simple (elementary) rate-controlling step. In fact, the decomposition is a complex chain reaction.²

Molecularity of a reaction refers to the number of molecules of reactants which are involved in the elementary chemical event. Molecularity pertains then to the molecular mechanism, while order pertains to the experimental rate equation. In the Ogg mechanism [Eq. (2-8)] the molecularities are 1, 2, and 2, respectively.

Only if the rate equation reflects the elementary molecular events will order and molecularity be equivalent. In the celebrated reaction



¹ R. A. Ogg, Jr., *J. Chem. Phys.*, **15**: 337, 613 (1947).

² S. W. Benson, "Foundations of Chemical Kinetics," p. 379, McGraw-Hill, New York, 1960.

detailed studies¹ indicate that the mechanism probably involves two HI molecules colliding and rearranging to produce H₂ and I₂. The experimental description of the decomposition (the rate equation) reveals second-order kinetics; i.e.,

$$-\frac{d(\text{HI})}{dt} = k(\text{HI})^2 \quad (2-11)$$

In this case order and molecularity coincide. Correspondence between order and overall stoichiometry is occasionally found but must be considered fortuitous, for example, $\text{NO} + \frac{1}{2}\text{O}_2$.

In sum, the experimentally determined order is not in principle uniquely related to overall stoichiometry, for example, $2\text{A} + \text{B} \rightarrow \text{P}$, so that one cannot expect a rate law of the form

$$\text{Rate} = kA^2B$$

As the overall expression or balance can be constructed from several differing elementary sequences (mechanisms), the form of the observed rate law conveys nothing unique concerning the real mechanism.

Example The velocity of the reaction $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ was observed by Bodenstein, who found that the irreversible-reaction-rate model is

$$\text{Rate} = k(\text{NO})^2(\text{O}_2)$$

Here order and stoichiometry coincide; yet later studies indicate that not one but at least two elementary steps are involved, the slowest of the two being bimolecular; i.e.,

Step 1: $\text{NO} + \text{O}_2 = \text{NO}_3$ fast equilibration

Step 2: $\text{NO}_3 + \text{NO} \longrightarrow 2\text{NO}_2$ slow

so that the rate-controlling step is expressed as

$$\text{Rate} = k_2(\text{NO}_3)(\text{NO})$$

But $\text{NO}_3 = K_1(\text{NO})(\text{O}_2)$, and therefore

$$\text{Rate} = k_2 K_1(\text{NO})^2(\text{O}_2) = k(\text{NO})^2(\text{O}_2)$$

If we assume an entirely different mechanism, say

Step 1: $\text{NO} + \text{NO} = (\text{NO})_2$ fast equilibration

Step 2: $(\text{NO})_2 + \text{O}_2 \longrightarrow 2\text{NO}_2$ slow

then $\text{Rate} = k_2(\text{NO})_2\text{O}_2$

but $(\text{NO})_2 = K_1(\text{NO})^2$, and so

$$\text{Rate} = k_2 K_1(\text{NO})^2(\text{O}_2)$$

as observed.

¹ C. A. Eckert, and M. Boudart, *Chem. Eng. Sci.*, **18**: 144 (1963).

Therefore it is demonstrated that (1) agreement between the observed rate law and stoichiometry does not prove that the overall reaction balance reflects an elementary (one-step) event and (2) more than one mechanistic sequence can be formulated which, with a shrewd choice of the rate-controlling step, will lead to the observed rate law.

The precise specification of the mechanism requires data other than the kinetic law, e.g., identification of intermediate NO_3 and/or $(\text{NO})_2$, etc. The kinetic law is thus a phenomenological *model* rooted in kinetic observables and transmits to the observer only what apparently occurs as witnessed on the gross laboratory scale. Inferences concerning the detailed molecular events are permissible only when supported by microscopic explorations which complement the kinetic investigation.

The Proportionality or Rate Coefficient

For a reaction $\text{A} + \text{B} \rightarrow \text{P}$ it has been indicated that analysis of laboratory rate data might suggest

$$-\frac{dA}{dt} = kA^\alpha B^\beta \quad (2-12)$$

where α and β are empirically determined. The proportionality coefficient k is a specific rate constant at constant temperature and catalyst concentration. If wall or surface effects are manifest, k will be a function of extent of surface, etc., and (at a constant temperature, catalyst concentration, wall area, etc.) α is determined by observing the rate-versus- A behavior in an excess of B, so that

$$-\frac{dA}{dt} = k'A^\alpha \quad k' = kB^\beta \quad (2-13)$$

while β is similarly fixed by rate-versus- B behavior in the presence of excess A. The next task is that of specifying the nature of k . If the system is homogeneous (reaction occurs entirely in one phase) and catalyst concentration is fixed or the system is noncatalytic, then, as suggested by Arrhenius, k should depend upon temperature in the following fashion:

$$k = \mathcal{A} \exp\left(-\frac{E}{RT}\right) \quad (2-14)$$

while Wilhelmy found

$$k = \mathcal{A} \exp\left(-\frac{b}{T}\right)$$

where $\mathcal{A}$ is called a *preexponential* or *Arrhenius factor* and E is the activation energy. Empirically, chemical-reaction-rate data can be compactly organized in terms of concentration (or pressure) and temperature by

$$-\frac{dA}{dt} = \mathcal{A}A^\alpha B^\beta \cdots \exp\left(-\frac{E}{RT}\right) \quad (2-15)$$

In summary, Eq. (2-15) presents a phenomenological description of reaction velocity in terms of laboratory observables (temperature, concentration of reactants, products, inerts, catalysts, reactor surface area, and its nature). Before looking into specific reaction types and methods of kinetic analysis, it is perhaps worthwhile to touch upon some of the theoretical foundations of chemical kinetics. A study of the theories which give rational support to the laboratory-founded rate expressions [such as Eq. (2-15)] should enhance our understanding of rate processes and consequently lead to a more intelligent use of kinetic data and rate laws.

2-2 THEORETICAL BASES OF CHEMICAL KINETICS

In an attempt to explain the effect of temperature upon the rate of sucrose inversion, Arrhenius¹ suggested that an equilibrium existed between inert and active sucrose molecules. The equilibrium concentration of active molecules would depend upon temperature as dictated by thermodynamics, i.e.,

$$K = (\text{const}) \left(\exp \frac{-\Delta H^\circ}{RT} \right)$$

and so $S \rightleftharpoons S^*$ (active sucrose)

$$K = \frac{S^*}{S} = (\text{const}) \left(\exp \frac{-\Delta H^\circ}{RT} \right) \quad (2-16)$$

Now

$$\text{Rate} = (\text{const})(S^*) = K'S$$

$$\text{Rate} = (\text{const})(S) \exp \frac{-\Delta H^\circ}{RT}$$

or
$$\text{rate} = \mathcal{A}S \exp \left(-\frac{E}{RT} \right) = kS \quad \text{where } E \equiv \Delta H^\circ$$

E was defined by Arrhenius as the difference in heat content between the active and inert reactant molecules. E has become known as an *energy of activation*,² while $\mathcal{A}$ is often termed the *frequency (of collision) factor*. The relationship between E , ΔH , and $\mathcal{A}$ will be made more explicit below. The Arrhenius equation has been verified by numerous experiments, and so

$$k = \mathcal{A} \exp \left(-\frac{E}{RT} \right) \quad \text{or} \quad \ln k = \ln \mathcal{A} - \frac{E}{RT} \quad (2-17)$$

A plot of $\ln k$ versus $1/T$ should give a straight line of slope $-E/R$ and intercept $\ln \mathcal{A}$.

While providing a method by which rate-temperature data can be rationally organized, the Arrhenius theory provides no basis for predicting E and $\ln \mathcal{A}$. It is understandable that, given the success of the Arrhenius relation, subsequent

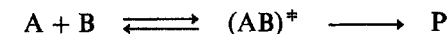
efforts would be devoted to more sophisticated interpretation of $\mathcal{A}$ and E . Inspired by developments of the kinetic theory of gases, the collision theory of chemical reaction emerged. The preexponential factor $\mathcal{A}$ was interpreted as a collision frequency Z . This collision frequency is (for at least two reactant molecules or atoms) determined by kinetic theory under limiting conditions.

Detailed treatments can be found in the usual physical chemistry texts. The predictive and instructive powers of collision theory are so limited that further elaboration is not justified. It suffices to note that experimental values of the Arrhenius coefficient $\mathcal{A}$ have been found to depart significantly from values suggested by collision theory. A more powerful model is set forth in the text of Glasstone, Laidler, and Eyring.¹ The essential features of this theory will now be considered.

Transition-State Theory

By *transition-state theory* we mean to emphasize the more telling aspects of *absolute-rate theory*, a term perhaps misleading in view of the very small number of even simple reactions whose absolute rates are subject to a priori prediction.

Somewhat in the spirit of Arrhenius' speculations cited above, transition-state theory asserts that in the reactants' progress along the path to products, an intermediate complex, or transition state, prevails; the transition-state complex exists in equilibrium with reactants; e.g., in the reaction $A + B \rightarrow P$, the theory states



Product (P) appears at a rate governed by the frequency of $(AB)^*$ decomposition in the forward direction, and, as will be shown below, the concentration of $(AB)^*$. Although $(AB)^*$ is not readily measured while A and B are detectable, the reactant-complex equilibrium can be stated in terms of thermodynamic activities of A , B , and $(AB)^*$:

$$K^* = \frac{a_{AB}}{a_A a_B} = \frac{\gamma_{AB} C_{AB}}{\gamma_A \gamma_B C_A C_B} \quad (2-18)$$

If it is assumed that the reaction rate is equal to the product of *concentration* of activated complex and the frequency of that complex decomposition, then

$$\text{Rate} = \nu \frac{\gamma_A \gamma_B}{\gamma_{AB}} K^* C_A C_B \quad (2-19)$$

Now $K^* = \exp(-\Delta G^*/RT)$, where ΔG^* is the free-energy difference between complex and reactants. Absolute-rate theory shows that the decomposition frequency ν is $k_B T/h$, where k_B is Boltzmann's constant, T is absolute temperature, and h is Planck's constant. Since $\Delta G^* = \Delta H^* - T \Delta S^*$,

$$\text{Rate} = \frac{k_B T}{h} \frac{\gamma_A \gamma_B}{\gamma_{AB}} C_A C_B \exp \frac{\Delta S^*}{R} \exp \frac{-\Delta H^*}{RT} \quad (2-20)$$

¹ S. Arrhenius, *Z. Phys. Chem.*, **4**: 226 (1889).

² We shall retain the symbol E and the term activation energy in view of the rather widespread use of the symbol and terminology. Actually activation enthalpy change ΔH is more precise.

¹ S. Glasstone, K. J. Laidler, and H. Eyring, "The Theory of Rate Processes," McGraw-Hill, New York, 1941.

but

$$\Delta H^* = E - RT + \Delta(PV)^* = E - RT + \Delta n RT = E + (\Delta n - 1)RT \quad (2-21)$$

Δn is the change in number of moles between reactants and activated complex:

$$\exp\left(\frac{-\Delta H^*}{RT}\right) = \exp\left(\frac{-E}{RT}\right) \exp(1 - \Delta n)^*$$

For an ideal-gas system, where $\gamma_A = \gamma_B = \gamma_{AB} = 1$,

$$\text{Rate} = \frac{k_B T}{h} \exp \frac{\Delta S_c^*}{R} \exp(1 - \Delta n)^* \exp\left(\frac{-E}{RT}\right) C_A C_B = k_c C_A C_B$$

Arrhenius suggested $k = \mathcal{A} \exp(-E/RT)$ or

$$\mathcal{A} = \frac{k_B T}{h} \exp \frac{\Delta S_c^*}{R} \exp(1 - \Delta n)^*$$

The a priori calculation of ΔS^* and ΔH^* requires knowledge of the nature and structure of the activated complex. Simple systems can be so described using the methods of statistical mechanics to compute K^* . Eckert and Boudart¹ illustrate the principles and the technique in the case of HI decomposition, where nonideality requires a priori determination of γ for reactants and the complex.

Relationship between Transition-State Theory and Arrhenius Model²

According to transition-state theory, in terms of pressure units,

$$\text{Rate} = \frac{k_B T}{h} K_p^* p_A p_B = k_p p_A p_B$$

$$\text{Then} \quad k_p = \frac{k_B T}{h} K^* = \frac{k_B T}{h} \exp \frac{\Delta S_p^*}{R} \exp \frac{-\Delta H^*}{RT} \quad (2-22)$$

Substituting Eq. (2-21) into (2-22) gives

$$k_p = \exp(1 - \Delta n)^* \frac{k_B T}{h} \exp \frac{\Delta S_p^*}{R} \exp \frac{-E}{RT} \quad (2-23)$$

$$\text{Therefore} \quad \mathcal{A} = \exp(1 - \Delta n)^* \frac{k_B T}{h} \exp \frac{\Delta S_p^*}{R} \quad (2-24)$$

where the rate is expressed in terms of pressure. In terms of concentration, since $k_c = k_p(RT)^{-\Delta n}$, where Δn is equal to the increase in number of molecules associated with creation of the activated complex from reactants,

$$k_c = \exp(1 - \Delta n)^* \frac{k_B T}{h} (RT)^{-\Delta n} \exp \frac{\Delta S_p^*}{R} \exp \frac{-E}{RT} \quad (2-25)$$

For example, for $A + B \rightleftharpoons (AB)^*$

$$\Delta n = -1 \quad \text{and} \quad (1 - \Delta n)^* = 2 \quad (2-26)$$

¹ Loc. cit.

² $\Delta S_p^* = \Delta S_c^* - \Delta n R \ln RT$, and $\exp(\Delta S_p^*/R) = \exp(\Delta S_c^*/R)(RT)^{-\Delta n}$.

we have

$$k_c = e^2 \frac{k_B T}{h} RT \exp \frac{\Delta S^*}{R} \exp \frac{-E}{RT}$$

Activity vs. Concentration

We have indicated that an implicit assumption in the above formulation of transition-state theory is that the rate is proportional to the *concentration* of the activated complex. This leads to the expression for k in the case $A + B \rightarrow (AB)^*$

$$k = \frac{k_B T}{h} \frac{\gamma_A \gamma_B}{\gamma_{AB}} \exp \frac{\Delta S^*}{R} \exp \frac{-\Delta H^*}{RT} \quad (2-27)$$

in the instance of nonideality of the reactant phase. If, on the other hand, the rate is proportional to the *activity* of the activated complex, then

$$k = \frac{k_B T}{h} \gamma_A \gamma_B \exp \frac{\Delta S^*}{R} \exp \frac{-\Delta H^*}{RT} \quad (2-28)$$

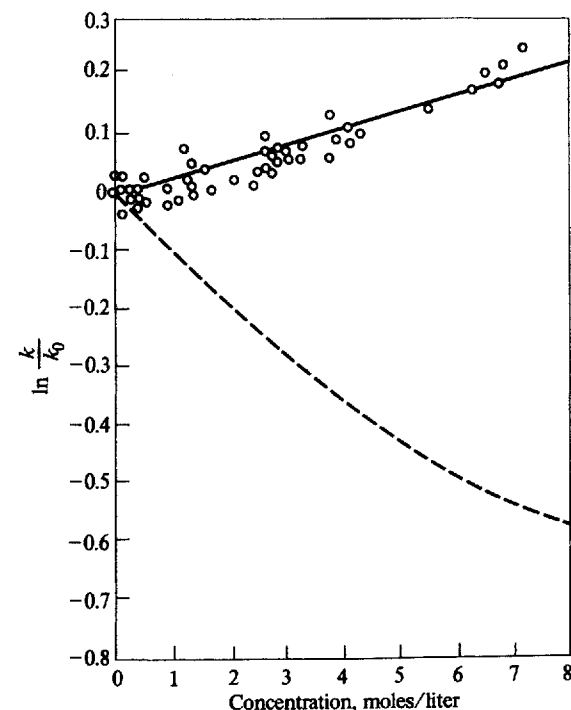


FIGURE 2-1

Variation in rate coefficient with total concentration for HI decomposition at 321.4°C. The solid line represents Eq. (2-27), and the dashed line represents Eq. (2-28). [C.A. Eckert and M. Boudart, *Chem. Eng. Sci.*, **18**:144 (1963).]

Eckert and Boudart¹ analyzed Kistiakowsky's rate data on HI decomposition as a function of concentration. If ideal-gas behavior prevailed, the second-order rate would vary with the square of concentration. Kistiakowsky's measurements revealed a greater effect of pressure than predicted by ideal-gas behavior. Eckert and Boudart's analysis of these data clearly demonstrates that it is the concentration rather than activity of the activated complex which dictates reaction rate [Eq. (2-27)]. Their paper illustrates the essential features of activated-complex formulation as well as the effects of nonideality upon the rate constant. Figure 2-1 shows the experimental and predicted effect of concentration on the specific rate constant.

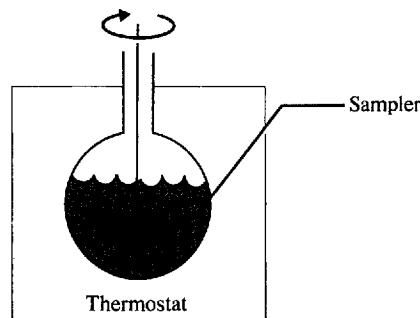
2-3 SOME DEFINITIONS OF REACTION ENVIRONMENTS

Laboratory Reactor Types

The general types of devices used to obtain raw kinetic data deserve mention at this point, as well as the terminology for describing the extent of reaction and product-generating efficiency.

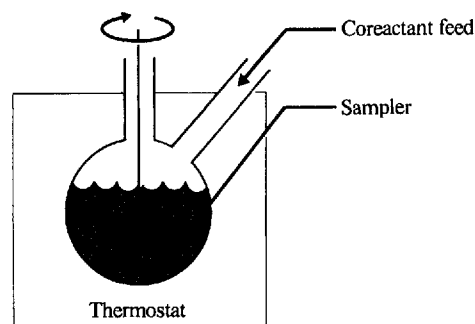
Laboratory-reaction studies are conducted in three general fashions:

1 Batch operation:



This is a closed system, and progress of reaction is monitored as a function of time t .

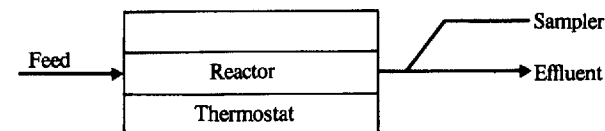
2 Semibatch operation:



One reactant is charged; the coreactant is added during the course of reaction.

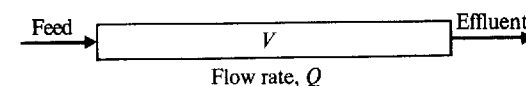
¹ Loc. cit.

3 Continuous-flow operation:



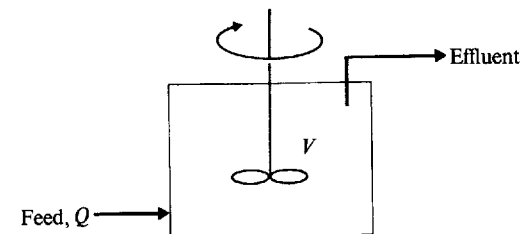
In contrast to batch and semibatch operation, continuous-flow reactors can be operated in steady state; i.e., the composition of effluent remains fixed with time if flow rate, temperature, and feed composition remain invariant. Reaction progress is followed as a function of residence time in the reactor rather than real time. This residence time (reactor volume divided by volumetric feed rate) can assume great importance depending upon the design of the flow reactor. Detailed discussion will be given in the next chapter; however, two limiting situations can be cited:

a Plug-flow reactor (PFR):



All molecules have same residence time V/Q , and concentrations vary only along the length of the tubular reactor.

b Continuous-flow stirred-tank reactor (CSTR):



Due to vigorous agitation, the reactor contents are well mixed, so that effluent composition equals that in the tank. As effluent and stirred-tank contents are of uniform composition, progress of reaction is monitored by noting the exit composition vs. average residence-time V/Q behavior in the steady state.

While quantitative discussions of type *a* and *b* flow reactors and intermediate types will appear in the next chapter, it is important to note that different information is acquired in securing reaction-progress-vs.-residence-time data from PFR and CSTR. In the PFR, since composition varies along the length of the tube, the effluent composition is an overall, integral composition. On the other hand, as the composition is uniform throughout the steady-state CSTR, the effluent-composition-vs.-average-residence-time data represent point or local

values. The PFR is a distributed-parameter unit (composition varies with position), while the CSTR is a lumped-parameter unit (uniform composition at all positions in the reactor). So for the CSTR, a steady-state material balance yields

$$QC_0 = QC + V\mathcal{R}$$

hence

$$\text{Rate, } \mathcal{R} = \frac{C_0 - C}{V/Q}$$

Laboratory Reactor Environments

The composition environment is implicitly noted above. In batch and semibatch operation uniformity throughout the geometric confines of the reaction volume must be realized at every moment during the course of reaction. Hence the necessity of vigorous agitation. Further, in a kinetic study the temperature field is of prime importance. Three general cases can exist:

- 1 *Isothermal* Heat is exchanged efficiently, so that at all times (or positions in a PFR) the temperature is constant.
- 2 *Adiabatic* Heat exchange between the reacting mass and the external surroundings is denied, either through design of the experiment, folly, or the inevitable consequence that the reaction rate (and thus heat generation or abstraction) is far too great to allow adequate exchange.
- 3 *Nonisothermal, nonadiabatic* A case more common than is generally acknowledged and one which is obviously intermediate between cases 1 and 2.

In the treatment of chemical-kinetic analysis which follows, isothermality is assumed in batch operation under conditions such that spatial composition gradients are absent. This means that compositions vary only with time, although the plug-flow case is noted to clarify the rigor which must be observed in expressing the differential reaction rate. Adiabatic operation on a laboratory scale is difficult to realize by design and although it is not as complex to analyze as nonadiabatic, nonisothermal data, it is preferable to follow laboratory procedures which generate isothermal data.

2-4 DEFINITIONS OF EXTENT OF REACTION

Conversion is expressed as the fraction of a key limiting reactant which is consumed to generate all reaction products. It is convenient to define conversion x as

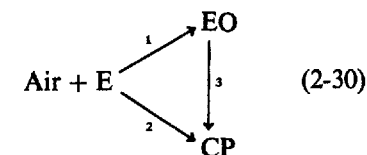
$$x = \frac{\text{moles reactant consumed}}{\text{moles of initial reactant}} \quad (2-29)$$

Yield can be defined in at least two ways: $Y(I)$ is moles of a particular product generated per mole of *initial* key reactant, and $Y(II)$ is moles of a particular product generated per mole of key reactant *consumed*.

Selectivity can be defined as moles of a particular product generated per mole of another product (by-product) generated. Selectivity is simply the ratio of two yields.

Conversion, yield, and selectivity are further specified as point (local) values and integral (overall) values.

These definitions can be clarified by example. Consider the oxidation of ethylene (E) in air to produce ethylene oxide (EO) and combustion products (CP) carbon dioxide and water. Assume the scheme



In a batch reactor

$$-\frac{d(E)}{dt} = k_1 f(E) + k_2 f'(E) \quad (2-31)$$

$$\frac{d(EO)}{dt} = k_1 f(E) - k_3 g(EO) \quad (2-32)$$

$$\frac{d(CP)}{dt} = k_2 f'(E) + k_3 g(EO) \quad (2-33)$$

where $f(E)$, $f'(E)$, and $g(EO)$ are general complex kinetic functions which need not be specified here. If E_0 is the initial ethylene concentration, conversion x is simply $(E_0 - E)/E_0$.

The distinction between point and overall conversion is not crucial. Overall conversion x_0 can be considered the final value at the end of reaction (or exit of a flow reactor), while the point value is clearly the intermediate value. In virtually all cases, one can expect conversion to increase with extent of exposure, though in an equilibrium reaction, an inadequate heat-removal policy can lead to conversion which passes through a maximum at some intermediate stage, e.g., in a nitrogen oxides-absorption tower.

Point yield of EO is simply the ratio of its rate of net appearance relative to that of reactant disappearance; thus

$$Y_p = \text{point or local yield} = \frac{d(EO)/dt}{d(E)/dt} = \frac{d(EO)}{d(E)}$$

Integral yield of EO is simply the result of integrating the point-value function. This gives either

$$Y_0(I) = \frac{EO}{E_0} = \text{function of } x, k_1, k_2, k_3, \text{ etc.} \quad (2-34)$$

or

$$Y_0(II) = \frac{EO}{E_0 - E} = \text{function of } x, k_1, k_2, k_3, \text{ etc.} \quad (2-35)$$

and so
$$\frac{EO}{E_0 - E} = \frac{EO}{E_0 x} \quad \text{or} \quad Y_0(II) = \frac{Y_0(I)}{x} \quad (2-36)$$

Point selectivity, as noted above, is a ratio of yields:

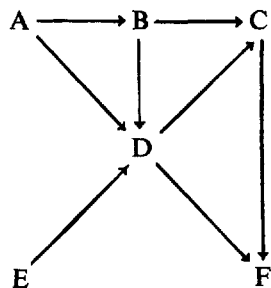
$$\mathcal{S}_p = \frac{Y_p(EO)}{Y_p(CP)} = \frac{d(EO)}{d(CP)}$$

Overall selectivity is secured by integration of $\mathcal{S}_p$, which in this example gives EO/CP as some function of conversion, etc. It must be borne in mind that the functions cited regarding integral yield and selectivity can be complex, and indeed in many important cases analytical integration is not possible. In terms of experimental realities, the integrated or overall yield and selectivity are, like conversion, the values found at the end of reaction, while point values are obviously those found at any given moment (or position in a PFR).

The above definitions of conversion, yield, and selectivity are not by any means unique or general.¹ In many quarters "conversion" to a product is accepted terminology for yield. In other instances yield and selectivity are used interchangeably.

2-5 MATHEMATICAL DESCRIPTIONS OF REACTION RATES

We shall now turn our attention to detailed mathematical descriptions of various reaction types. Chemical reactions may be simple, for example, $A \rightarrow B$, or complex



In addition to the inherent complexity illustrated in the latter scheme, mixed orders, reversibility, and volume changes due to stoichiometry conspire to complicate solution of the differential equations describing the system. Our descriptions begin with a definition of the rate of chemical reaction.

Reaction rate is defined as the change in moles n of a component with respect to time per unit of reaction volume, i.e.,

$$\text{Rate} = \frac{1}{V} \frac{dn}{dt} = \text{function}(A, B, C, \dots, \text{temperature}) \quad (2-37)$$

The expression assumes that a batch system exists, so that time t is the time of retention within a closed vessel (of variable or fixed volume). For fixed volume, batch ($n = CV_0$ = number of moles):

$$\begin{aligned} \text{Rate} &= \frac{1}{V_0} \frac{dn}{dt} \\ &= \frac{1}{V_0} \frac{dC V_0}{dt} = \frac{dC}{dt} \quad \begin{array}{l} V = V_0, C = C_0 \text{ at } t = 0 \\ V = V_0, C = C \text{ at } t > 0 \end{array} \end{aligned} \quad (2-38)$$

and for variable volume, batch ($n = CV$):

$$\begin{aligned} \text{Rate} &= \frac{1}{V} \frac{dn}{dt} \\ &= \frac{dC}{dt} + \frac{C}{V} \frac{dV}{dt} \quad \begin{array}{l} V = V_0, C = C_0 \text{ at } t = 0 \\ V = V \neq V_0, C = C \text{ at } t > 0 \end{array} \end{aligned} \quad (2-39)$$

Suppose, on the other hand, a simple isothermal tubular reactor (PFR) is being fed at a constant rate F_0 , moles per time. A balance over a differential volume dV

$$F - (F + dF) = (\text{rate}) dV$$

$$\text{gives} \quad -\frac{dF}{dV} = \text{rate}$$

Two cases may be encountered in such a system: (1) there is no change in moles due to reaction, or (2) a mole change due to reaction exists, in which case the volumetric flow Q will vary with extent of reaction (distance along the PFR). In case 1, at constant temperature and pressure, the volumetric flow rate Q remains constant along the length of the reactor. In case 2, Q changes as the number of moles change due to reaction.

CASE 1 There is no mole change, and $Q = Q_0$:

$$\text{Rate} = \frac{dF}{dV} = \frac{d(QC)}{dV} = Q_0 \frac{dC}{dV} = \frac{dC}{d\tau} \quad \tau = \frac{V}{Q_0} \quad (2-40)$$

Comparing this result to the fixed-volume batch case, we see that contact time $\tau = V/Q_0$ in a flow system is equal to holding time t in the fixed-volume batch system *only* if Q_0 refers to reaction temperature and pressure.

CASE 2 There is mole change due to reaction, that is, $Q \neq Q_0$:

$$\text{Rate} = \frac{dF}{dV} = Q \frac{dC}{dV} + C \frac{dQ}{dV} \quad (2-41)$$

In this second case, the change in flow rate due to reaction as a function of volume traversed must be specified, just as the change in reaction volume with time must be specified in the batch variable-volume case.

¹ H. H. Voge, Letter to Editor, *Chem. Eng. News*, Feb. 21, 1966.

The merits of using conversion x , defined earlier, become evident in formulating the variable-volume reaction scheme. As noted above, for the PFR, in terms of concentration

$$\frac{dF}{dV} = Q \frac{dC}{dV} + C \frac{dQ}{dV} \quad (2-42)$$

However, as

$$x = \frac{\text{moles of reactant consumed}}{\text{moles of reactant in feed}}$$

for F_0 = moles reactant fed per time

$$F = F_0 - xF_0 = F_0(1 - x) \quad \text{and} \quad F_0 = Q_0 C_0$$

$$\text{and so} \quad \frac{dF}{dV} = -F_0 \frac{dx}{dV} = \text{rate} = -C_0 \frac{dx}{d\tau_0} \quad \text{where } \tau_0 = \frac{V}{Q_0} \quad (2-43)$$

a far simpler expression than that in terms of concentration.

By similar reasoning, for a batch case, since $n = n_0(1 - x)$,

$$\frac{1}{V} \frac{dn}{dt} = -\frac{n_0}{V} \frac{dx}{dt} = \text{rate} = -C_0 \frac{dx}{dt} \quad (2-44)$$

Note that in each case (PFR or batch) the rate functionality must be expressed in terms of conversion x to facilitate integration (analytical or otherwise). In this chapter we shall largely confine ourselves to constant-volume batch or flow systems without mole change due to reaction. We justify this restriction at this stage because our intent here is to present integrated forms of rate equations which can be profitably used in interpreting laboratory data usually secured in batch systems of constant volume or in flow systems where reactant dilution with inerts or low conversion is frequently realized to assure a negligible total flow change due to reaction.

2-6 CLASSIFICATION OF REACTIONS

One may categorize reactions by order, reversibility, complexity, whether homogeneous or heterogeneous, and catalyzed or not. For example, ethylene is hydrated with water in the presence of dissolved acid catalysts to produce ethyl alcohol. The reaction is obviously catalytic. At high pressure ethylene gas dissolves and reacts in liquid water (containing H_2SO_4) to yield the alcohol. This is a heterogeneous system (gas-liquid); the reaction occurs in one phase (homogeneous) and is therefore catalyzed in that phase (homogeneous catalysis).

On the other hand, ethylene gas and H_2O vapor will react to give alcohol when passed through a tube packed with solid particles containing acidic properties. This is a heterogeneous system (gas-solid); the reaction occurs at the gas-solid interface (heterogeneous) and is solid-catalyzed (heterogeneous catalysis).

In this chapter we shall deal explicitly with kinetics of reactions in one phase (homogeneous). Implicitly homogeneous catalysis is manifest in the rate constant k , as we noted earlier in discussing sucrose inversion. Kinetics of heterogeneous catalysis will be treated in Chap. 8.

For constant-volume batch and tubular-flow reactions with negligible volume change due to reaction, the following topics will be treated:

- 1 Simple reactions
 - a First, second, third, and n th order, irreversible
 - b Reversible cases for some simple reactions
 - c Relationship between equilibrium and rate constants
- 2 Complex reactions
 - a Simultaneous reactions
 - b Higher-order simultaneous reactions
 - c Concurrent and consecutive linear reactions
 - d Higher-order consecutive reactions
 - e The general complex linear network
 - f Autocatalysis, chain reactions, explosions, and simple polymerization

2-3 SIMPLE REACTIONS

Preliminary Remarks

In a general case $aA + bB \cdots \rightarrow pP + qQ \cdots$ the reaction rate (at constant volume) is expressed in terms of concentrations A , B , etc., and stoichiometric coefficients as

$$\text{Rate} = -\frac{1}{a} \frac{dA}{dt} = -\frac{1}{b} \frac{dB}{dt} = \frac{1}{p} \frac{dP}{dt} = \frac{1}{q} \frac{dQ}{dt}$$

For example in the gas-phase reaction $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$

$$\frac{d(\text{NO})}{dt} = 2 \frac{d(\text{O}_2)}{dt} = -\frac{d(\text{NO}_2)}{dt}$$

Since $d(\text{NO})/dt = -k_{\text{NO}}(\text{NO})^2(\text{O}_2)$, if oxygen disappearance is measured,

$$\frac{d(\text{O}_2)}{dt} = -k_{\text{O}_2}(\text{NO})^2(\text{O}_2) = -\frac{k_{\text{NO}}}{2}(\text{NO})^2(\text{O}_2)$$

In terms of appearance of product NO_2

$$\frac{d(\text{NO}_2)}{dt} = k_{\text{NO}_2}(\text{NO})^2(\text{O}_2) = k_{\text{NO}}(\text{NO})^2(\text{O}_2)$$

Note that the rate of NO oxidation is second order in NO, first order in O_2 , and zero order in NO_2 . Overall order is third. The orders found experimentally may suggest to the novice a reaction involving three-body collision (molecularity of 3). Actually the reaction is complex,¹ as noted above.

¹ J. J. Carberry, *Chem. Eng. Sci.*, 9: 189 (1959).